# CAVA: Context-Aware Video Adaptation for AV Teleoperation

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## 1 Motivation

Remote vehicle operation requires **fresh, clear video from multiple cameras** so the operator can understand the surroundings and choose a maneuver.

### High demand from multi-camera video

Simultaneous feeds cover the road ahead, adjacent lanes, intersections, and blind spots. Preserving useful detail and frequent updates across every camera creates a large aggregate video uplink load.
Average combined throughput requirement can exceed **30 Mbps.**

### Low and variable uplink capacity

Commercial 5G uplink capacity is much lower than downlink capacity. Mobility, blockage, fading, contention, and handovers cause sudden capacity drops. When video demand exceeds capacity, queues build and frames become stale or get dropped.

**More transmitted video does not guarantee timely, useful information.**

## 2 Key idea: not all bits are equal

**With each new set of frames, different camera views and different components within those frames matter. Their importance depends on what the operator needs to do and the current situation.**

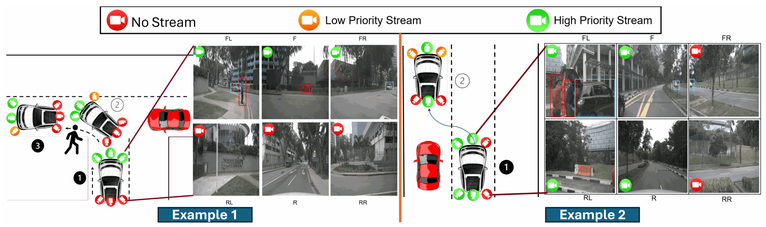



Fig. 2. A left turn and a lane change require different views. Relevance changes with the intended maneuver and surrounding actors.

### What does the operator need to do?

Turning, proceeding straight, and merging require different parts of the surround view. For a left turn, views that expose approaching traffic and the intended path matter more than an empty rear view. **Intent** determines which information supports the next decision.

### What is happening now?

An approaching vehicle or pedestrian can make a view newly important. Rapid motion needs frequent updates, while rich scenes need spatial detail. Within a frame, relevant road users, traffic controls, and road regions deserve more bits than background.

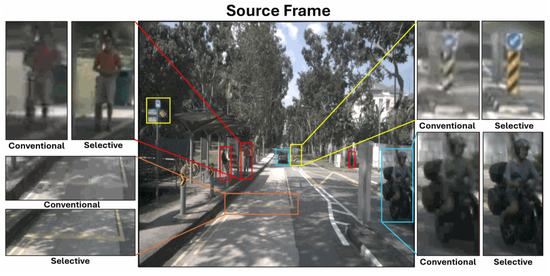



Fig. 4. Uniform and selective quantization at the same 500 kbps budget. Preserving relevant regions can retain useful detail while degrading background.

## 3 Architecture: joint context

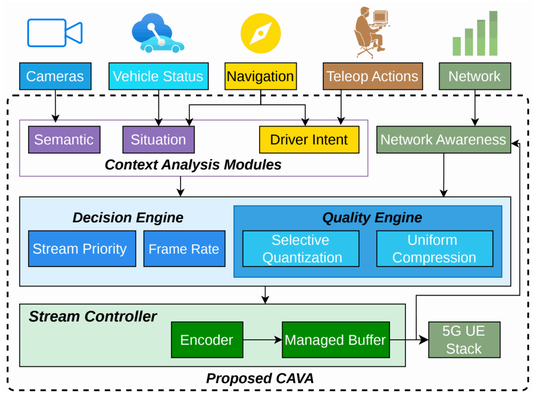



Fig. 6. Context analysis informs package-wide decisions, which the encoder and stream controller enforce.

Assess surroundings: semantic masks locate relevant regions, while situational analysis measures richness, motion, and actor relevance. Infer the maneuver from navigation and operator actions. Estimate the available network capacity.

**Use this joint context to rank parts of the camera package and choose active views, frame rates, and region quality under the shared uplink budget. The controller prioritizes fresh frames and limits stale backlogs.**

## 4 Evaluation methodology

### CROC and CRIF: task-relevant delivery

Context-Relevant Object Coverage (CROC): fraction of unique relevant objects delivered and matched within a latency bound, averaged across time windows.

Context-Relevant Information Fidelity (CRIF): object-region SSIM for relevant objects delivered on time. Interpret with CROC because fidelity is conditional on delivery. An offline oracle supplies relevance labels for evaluation.

### Trace-based and real 5G evaluation

850 nuScenes scenes, each 19–21 s, with six synchronized cameras. Mahimahi replays low-CQI and RB-contention traces against WebRTC-GCC and SCReAM. Commercial urban 5G compares CAVA with GCC. All use matched VP9 settings and hardware.

### Human study A: feed preference

Participants first view the complete ground-truth feed, then anonymized system outputs in randomized order under the same scene and trace. Each trial selects one preferred feed. This measures perceived utility, independently of timeliness.

### Human study B: usable-fidelity thresholds

Participants judge 25 fixed target frames selected from a 20-scene catalog, viewing progressively degraded versions until a target becomes unacceptable. The last accepted object-region SSIM yields a fidelity floor. Studies A and B use disjoint scenes.

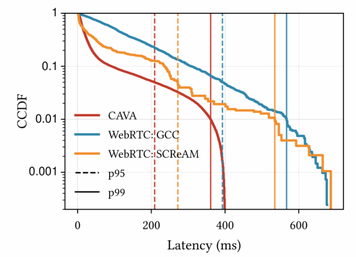
### TAP: Time to Actionable Perception

Use the human-derived class thresholds to measure time from an object's first physical appearance to receipt of its first usable observation. TAP connects network delay, object coverage, and fidelity. Never-usable tracks remain in the CDF denominator.

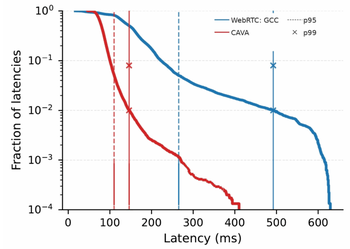
**CROC asks whether relevant objects arrive on time. CRIF asks how clearly they arrive. TAP asks how soon they become usable to a human.**

## 5a Results: trace replay and real 5G

### Trace replay: network latency




### Commercial 5G: end-to-end




Figs. 9a and 13. CAVA lowers latency under adverse trace replay. On commercial 5G, mean end-to-end latency falls from 157 to 82 ms and p99 from 492 to 146 ms.

### Trace replay: relevant-object coverage and fidelity

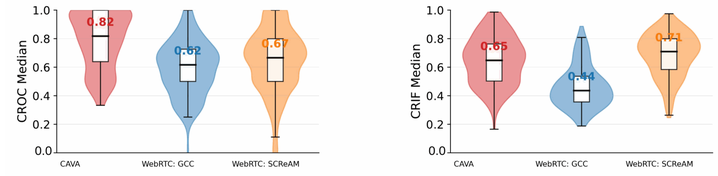



Fig. 9b–c. Median CROC: 0.82 / 0.62 / 0.67. Median CRIF: 0.65 / 0.44 / 0.71 for CAVA / GCC / SCReAM. SCReAM's higher conditional fidelity accompanies lower coverage.

## 5b Results: human-grounded evaluation

Human judgments establish both feed utility and the fidelity thresholds used to measure actionable perception.

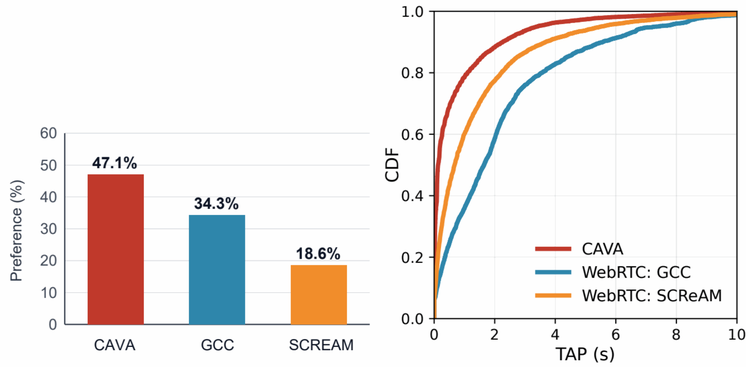



Fig. 17. Feed preference (left) and cumulative fraction of relevant tracks reaching human-usable fidelity over time (right).

### Feed preference

CAVA: 47.1% of trials, GCC: 34.3%, SCReAM: 18.6%.

### Human-derived object SSIM floors

Vulnerable road users: 0.82. Cars and large vehicles: 0.80. These class medians define when an observation becomes usable in TAP.

### Time to Actionable Perception

Median TAP: 0.13 s for CAVA, 0.68 s for SCReAM, and 1.65 s for GCC. CAVA reduces median TAP by 81% and 92%, respectively.

Source: CAVA paper, Sections 1, 3–6 and Appendix F. Original paper figures. Oracle relevance is for offline evaluation. Human fidelity floors are a representative calibration, not a driving-safety validation.